2A1 Time Frequency Analysis: Fourier Series and Transforms

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Course Page	www.robots.ox.ac.uk/~dwm/Courses/2TF

Grey Book

Complex Fourier series: evaluation of complex coefficients for periodic functions; inversion relationship; the idea of spectra.

Fourier transform: derivation of transform from Fourier series; inverse transform; convolution integral; impulse response functions; proof and use of duality; convolution and Parsevals theorems.

Introduction to sampling and reconstruction, including the sampling theorem and aliasing.

Introduction to random processes.

Lecture Content

1. From Signals to Complex Fourier Series
2. From Complex Fourier Series to the Fourier Transform
3. Convolution. The impulse response and transfer functions
4. Sampling, Aliasing
5. Power & Energy Spectra, Autocorrelation, and Spectral Densities
6. Random Processes and Signals

The content will flow from one lecture slot to another. If there is any time left over, we will go through examples.

References

These notes are not meant to be comprehensive. Fourier Analysis is a topic where a good book with decent diagrams and examples can make a difference. So — do read promiscuously.

This list will be updated on the website with comments.

- Kreyszig, *Advanced Engineering Mathematics 6th ed*, Wiley
- Riley, Hobson and Bence, *Mathematical Methods for the Physics and Engineering*, CUP.
- Baher *Analog and Digital Signal Processing* Wiley, 1991.
- Denbigh, *System Analysis and Signal Processing: with emphasis on the use of MATLAB*, Addison-Wesley, 1998.
- James, *Advanced Modern Engineering Mathematics* Addison-Wesley, 1993.
- Oppenheim, Willsky and Young, *Signals and Systems*, Prentice-Hall, 1984.
- Newland *Introduction to Random Vibrations, Spectral and Wavelet Analysis*, Longman, 1993.
- Papoulis *Probability, Random Variables and Stochastic Processes* McGraw-Hill, 1991.

Course WWW Pages

Pdf copies of these notes (including larger print versions), pdf copies of the lecture slides, the tutorial sheets, FAQs etc will be accessible from

www.robots.ox.ac.uk/∼dwm/Courses/2TF

Just the notes and the tute sheets get put on weblearn.

Topic 1

From Signals to Complex Fourier Series

1.1 Linear systems and frequency

By this stage, you will have realized how important *linear systems* and their analysis are to engineers. In a linear system — one described by a linear differential equation of some order — the response $y(t)$ occurs at the same frequency as the input $x(t)$, and if the input's amplitude is changed by some factor, the output's amplitude will change by that same factor. When faced with a non-linear system, engineers often *linearize the system* by considering incremental inputs and outputs that occur around a fixed operating point. The use of small signal equivalents in transistor circuits is an obvious example.

1.1.1 You know about Frequency Response

You will have an understanding too of the importance of the frequency response of a system, be it mechanical, electrical, or whatever. That response is described by the *transfer function* $H(\omega)$, which may be complex, representing both the change in phase and amplitude between input and output. Strictly speaking, the transfer function relates inputs and outputs in the frequency domain

$$Y(\omega) = H(\omega)X(\omega) \ ,$$

but so far you have thought of $X(\omega)$ and $Y(\omega)$ as phasor representations of harmonic input and output at a single frequency. At a single frequency, the temporal input and output are related to the frequency representations by

$$x(t) = \mathrm{Re}\left(X(\omega)\mathrm{e}^{\mathrm{i}\omega t}\right) \ ; \qquad y(t) = \mathrm{Re}\left(Y(\omega)\mathrm{e}^{\mathrm{i}\omega t}\right) = \mathrm{Re}\left(H(\omega)X(\omega)\mathrm{e}^{\mathrm{i}\omega t}\right)$$

As a concrete example, consider as input the voltage $x(t) = V_0 \cos\omega t$ applied to an inductor L in series with a resistor R, and as output the voltage $y(t)$ across R.

The input phasor, transfer function and output phasor are, respectively

$$X(\omega) = V_0 \; ; \quad H(\omega) = \frac{R}{R + \mathrm{j}\omega L} \; ; \quad Y(\omega) = \frac{V_0 R}{R + \mathrm{j}\omega L} \; ;$$

Notice that although we can write $Y(\omega) = H(\omega)X(\omega)$ there seems to be no equivalently crisp operation involving $y(t)$ and $x(t)$.

1.1.2 You know about Superposition

Linear systems obey the *principle of linear superposition*. It says that if the input is a linear combination of signals, the output is the same linear combination of the individual outputs

$$x(t) = \alpha_1 x_1(t) + \alpha_2 x_2(t) + \ldots \quad \Rightarrow \quad y(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t) + \ldots \; .$$

If the $x_1(t)$ etc are harmonic signals at frequencies ω_1 etc, the output must be

$$y(t) = \alpha_1 \mathrm{Re}\left(H(\omega_1)X(\omega_1)\mathrm{e}^{\mathrm{i}\omega_1 t}\right) + \alpha_2 \mathrm{Re}\left(H(\omega_2)X(\omega_2)\mathrm{e}^{\mathrm{i}\omega_2 t}\right) + \ldots \; .$$

A rather more elegant way of thinking about this is to write the discrete frequency spectrum of $x(t)$ as

$$X(\omega) = \alpha_1 X(\omega_1) + \alpha_2 X(\omega_2) + \cdots$$

then

$$Y(\omega) \;=\; \alpha_1 Y(\omega_1) + \alpha_2 Y(\omega_2) + \cdots \;=\; \alpha_1 H(\omega_1)X(\omega_1) + \alpha_2 H(\omega_2)X(\omega_2) + \cdots \; .$$

1.1.3 You know about Fourier Series

This is remarkable, but would be utterly arcane were it not for an amazing property of (most) periodic signals, viz:

> A periodic signal of angular frequency ω_0 can[1] be represented as the sum of a set of harmonic signals at frequencies ω_0, $2\omega_0$, $3\omega_0$, and so on.

These sums of harmonic waves are *Fourier Series*. For example, the Fourier series of a unit square wave with a zero at $t = 0$ and period $T = 2\pi/\omega_0$ is

$$f(t) = \frac{4}{\pi}\left[\sin\omega_0 t + \frac{1}{3}\sin 3\omega_0 t + \frac{1}{5}\sin 5\omega_0 t + \cdots\right] = \frac{4}{\pi}\sum_{n \text{ odd}} \frac{1}{n}\sin n\omega_0 t \; .$$

Now, armed with a system's transfer function $H(\omega)$, the principle of linear superposition, and this and similar Fourier Series, you can work out the output of the system corresponding to the square wave, or any other periodic input. This is illustrated in Figure 1.1.

[1] Provided the "Dirichlet conditions" are satisfied. See later.

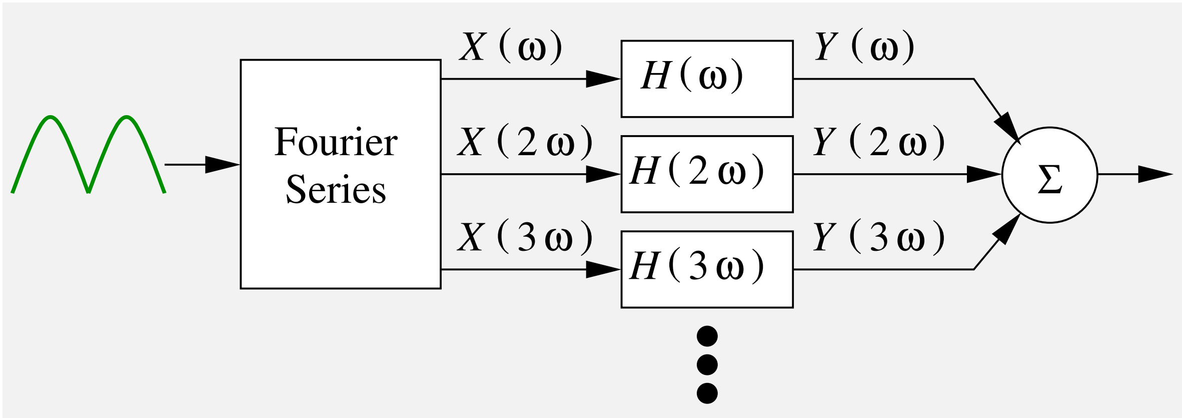


Figure 1.1: The Fourier Series, the principle of linear superposition, and the transfer function, allow one to compute the output for any periodic input.

1.2 The gap in our knowledge

Unfortunately, not all inputs are periodic. Can Fourier analysis help us? Well, if the signal $x(t)$ is of finite duration we might make it periodic by "pretending" it repeats. However, this does not help with general signals of infinite duration.

Another way of perceiving the gap in out knowledge is to realize that *well-behaved periodic functions give rise only to discrete frequency spectra*. For example, the unit square wave (Figure 1.2(a)) has the frequency spectrum shown in Figure 1.2(b), where the components are in the ratio $1 : (1/3) : (1/5) : (1/7) : \ldots$. However, instinct tells us that there must be signals that have a continuous frequency spectrum as sketched in Figure 1.2(c). Nothing we know *yet* about Fourier analysis would allow the analysis of a continuous spectrum.

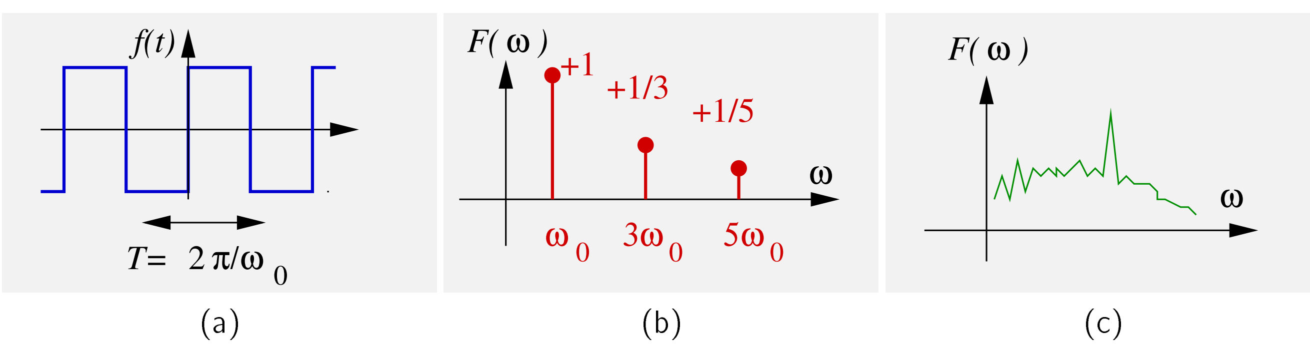


Figure 1.2: A periodic function (a) gives a discrete frequency spectrum (b). (c) A continuous frequency spectrum cannot be derived from a periodic function.

1.3 Fourier Transforms

We shall see that *Fourier transforms* provide a method of transforming infinite duration signals, both non-periodic and periodic, from the time domain into the *continuous frequency domain*.

In fact, they provide an entire language with which to work and think in the frequency domain. The language involves a deal of new vocabulary and several new mathematical techniques, like *convolution*, *correlation*, *modulation*, *sampling*, *spectral density*, *δ-functions* and so on.

Most of these techniques depend at the lowest level on integration. The integrals can look daunting, but it is important to rise up onto the next level — so that you can say "a signal's power spectral density is the Fourier transform of its auto-correlation". Success will come if you practice the mathematics, but also practice fixing the concepts in your head by using simple physical examples.

1.4 Joseph Fourier

If at any point you become really cross, just remember that Fourier seems a really jolly chap. He had an interesting life not only in mathematics but in politics in France during Napoleon's time. Read his biography (and those of many other mathematicians) at the website maintained by the Department of Maths and Computer Science at St Andrew's University. Follow the links from www-history.mcs.st-and.ac.uk/∼history/

1.5 Signals

You will have noticed that the emphasis of our introductory discussion drifted from *systems*, and towards *signals*. Indeed, this course could be titled "An introduction to analogue signal processing". Let us start by defining various signal types.

A signal might be a function of one or of several variables. Space and time are very common variables, but here we will tend to stick to one variable, and choose time t more often than not. Again more often than not, we will think about electrical signals — but do remember that the variation in temperature during the day is just as much a signal as the voltage output of a thermocouple sensing it.

An Analogue signal is one whose amplitude covers a continuous range. It may be bounded (e.g. $0-5$ V) but it can just as easily take the value 2.0572941675975 V as 2.0572941675974 V, and indeed anything between!

A continuous-time analogue signal is one that has a value for a continuous sweep of values of its parameter, t. Some examples are shown in Figure 1.4. Notice the

Figure 1.3: Jean-Baptiste Joseph Fourier 1768-1830.

value can be zero, but it is defined as zero; and notice too that a continuous-time signal does not have to be a continous function.

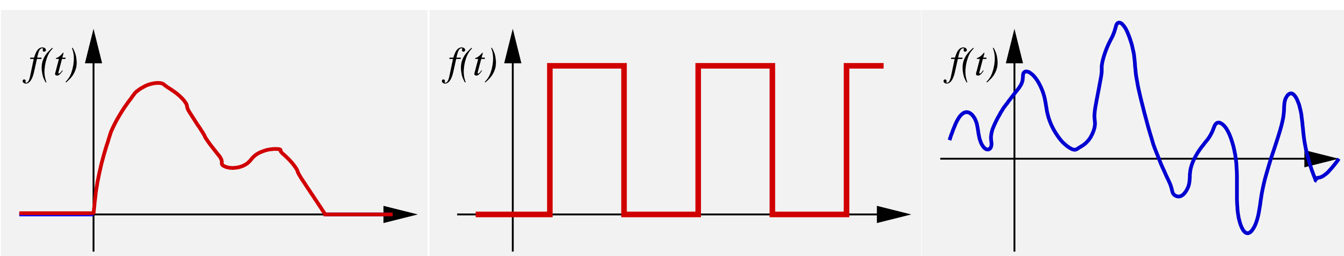
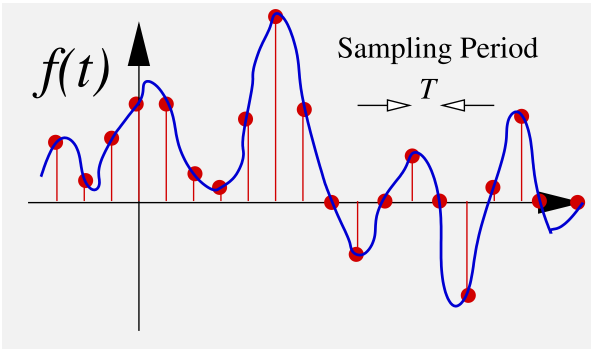


Figure 1.4: Examples of continuous-time analogue signals.

A discrete-time analogue signal is one that has an analogue value at certain times only. Typically these will be at regular intervals, arising from regular sampling. Examples are shown in Figure 1.5. Note that rather than $f(t)$, this type of signal is labelled $f(nT)$, where n is an integer, and T is the sampling interval.

A digital signal is one that is by definition sampled, but whose amplitude can only take one of a discrete set of values represented by some binary coding scheme. Suppose we used 4 bits. This could represent $\{0, 1, \ldots, 15\}$ V, or $\{-0.2, -0.1, 0.0, \ldots, 1.3\}$ V, or $\{0, 1, 4, 9, \ldots, 225\}$ V; but, using the first example, we cannot properly represent any value between 0 and 1 V. In this course we are not concerned with digital signals, and consider only analogue, continuous-time and discrete-time.



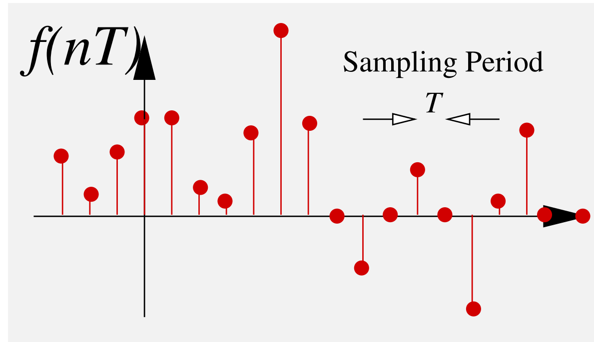


Figure 1.5: A continuous time signal $f(t)$ sampled at intervals of T to general a discrete-time signal $f(nT)$.

A causal signal is one that is finite only for $t > 0$ — ie, $f(t) = 0$ for all $t < 0$.

A casual signal is one that has been mis-spelled. See above.

A deterministic signal is one that can be described by a function, mapping, or some other recipe or algorithm. If you know t, you can work out $f(t)$. We shall be interested in deterministic signals for much of the course.

A random signal is determined by some underyling random process. Although its statistical properties might be known (e.g. you might know its mean and variance) you cannot evaluate its value at time t. You might by able to say something about the *likelihood* of its taking some value at time t, but not more. We will come to think about random processes towards the end of the course.

1.6 Orthogonal basis functions - Revision

Before revising the Fourier series, we think about orthogonality and functions, taking a scenic meander via vectors.

Consider three vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, which are of different lengths but lie at right angles to each other. They form a set of orthogonal basis vectors in 3D. It should be evident that any 3D vector $\mathbf{f}$ can be described by a unique linear combination of the vectors

$$\mathbf{f} = A_1\mathbf{v}_1 + A_2\mathbf{v}_2 + A_3\mathbf{v}_3 \ .$$

How would you find these unique coefficients for a particular $\mathbf{f}$? To find A_1, take the scalar or inner product of both sides with $\mathbf{v}_1$

$$\mathbf{f} \cdot \mathbf{v}_1 = A_1\mathbf{v}_1 \cdot \mathbf{v}_1 + A_2\mathbf{v}_2 \cdot \mathbf{v}_1 + A_3\mathbf{v}_3 \cdot \mathbf{v}_1 \ ,$$

and then exploit orthogonality which tells you that $\mathbf{v}_2 \cdot \mathbf{v}_1 = 0$ and $\mathbf{v}_3 \cdot \mathbf{v}_1 = 0$, so that

$$A_1 = \frac{\mathbf{f} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \quad .$$

(The denominator is important — $\mathbf{v}_1$ and so on were not unit vectors.)

In the early 19th century it was realized that it was possible to treat functions f in the same way and to make them up from a set of orthogonal basis *functions*. Pretend there is a set of orthogonal "v-functions", so that, similar to the vector case,

$$f(t) = A_1 v_1(t) + A_2 v_2(t) + A_3 v_3(t) + \cdots \; .$$

How do we find A_1 and so on? First we need to define the equivalent of the scalar or inner product between functions — but for now just assume we can, and we denote it $\langle f, g \rangle$. Then, by analogy

$$\langle f, v_1 \rangle = A_1 \langle v_1, v_1 \rangle + A_2 \langle v_2, v_1 \rangle + \cdots \quad .$$

But $\langle v_2, v_1 \rangle = 0$ and so on, so that

$$A_1 = \frac{\langle f, v_1 \rangle}{\langle v_1, v_1 \rangle} \qquad \text{and} \qquad A_n = \frac{\langle f, v_n \rangle}{\langle v_n, v_n \rangle} \; .$$

There are a considerable number of famous sets of orthogonal basis functions, many derived by French mathematicians, but it was Fourier who noticed that the cosines and sines made up such a set which were then capable of representing periodic functions.

1.7 Fourier Series

Fourier found that a periodic function $f(t)$, with period T, can be written as a sum of cosine and sine functions of the fundamental frequency $\omega = 2\pi/T$ and its harmonics 2ω, 3ω, etc

$$\boxed{f(t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega t) + \sum_{n=1}^{\infty} B_n \sin(n\omega t) \quad .}$$

The expressions for A_n and B_n are found from the orthogonality conditions as

$$\boxed{\begin{aligned} A_n &= \frac{2}{T}\int_{-T/2}^{+T/2} f(t)\cos(n\omega t)\mathrm{d}t \quad n = 0, 1, \ldots \Rightarrow \; A_0 = \frac{2}{T}\int_{-T/2}^{+T/2} f(t)\mathrm{d}t \\ B_n &= \frac{2}{T}\int_{-T/2}^{+T/2} f(t)\sin(n\omega t)\mathrm{d}t \quad n = 1, 2, \ldots \end{aligned}}$$

1.7.1 Finding the coefficients

To derive the stated expressions for the coefficients, we must first define the inner product for these functions. It is defined by the integral over a period, divided by the period.

Then, we should demonstrate that the basis functions, $c_n = \cos n\omega t$ and $s_n = \sin n\omega t$, are orthogonal. In other words, we must show that the inner products $\langle c_n, c_m\rangle$ and $\langle s_n, s_m\rangle$ are zero for $m \neq n$, that $\langle c_n, s_m\rangle = 0$, and that only $\langle c_n, c_n\rangle$ and $\langle s_n, s_n\rangle$ are finite.

One finds

$$\begin{aligned} \frac{1}{T}\int_{-T/2}^{T/2}\cos m\omega t\cos n\omega t\mathrm{d}t &= \begin{cases} 0 & m \neq n \\ 1/2 & m = n > 0 \\ 1 & m = n = 0 \end{cases} \\ \frac{1}{T}\int_{-T/2}^{T/2}\sin m\omega t\sin n\omega t\mathrm{d}t &= \begin{cases} 0 & m \neq n \\ 1/2 & m = n > 0 \\ 0 & m = n = 0 \end{cases} \\ \frac{1}{T}\int_{-T/2}^{T/2}\cos m\omega t\sin n\omega t\mathrm{d}t &= 0 \end{aligned}$$

The only complication is that we have both c_n and s_n as in the basis set, so we need two coefficients with the subscript n. Calling these A_n and B_n we find

$$A_n = \frac{\langle f, c_n\rangle}{\langle c_n, c_n\rangle} = \frac{\frac{1}{T}\int_{-T/2}^{T/2} f(t)\cos n\omega t\mathrm{d}t}{\frac{1}{T}\int_{-T/2}^{T/2}\cos n\omega t\cos n\omega t\mathrm{d}t} = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos n\omega t\mathrm{d}t$$

and

$$B_n = \frac{\langle f, s_n\rangle}{\langle s_n, s_n\rangle} = \frac{\frac{1}{T}\int_{-T/2}^{T/2} f(t)\sin n\omega t\mathrm{d}t}{\frac{1}{T}\int_{-T/2}^{T/2}\sin n\omega t\sin n\omega t\mathrm{d}t} = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\sin n\omega t\mathrm{d}t$$

1.7.2 A more explicit approach

If you felt there was sleight of hand in the above, you may prefer to write down the series, and then, to find the B_m for example, multiply it by $\sin(m\omega t)$ and average over a period:

$$\frac{1}{T}\int_{-T/2}^{T/2} f(t)\sin(m\omega t)\mathrm{d}t = \frac{1}{T}\int_{-T/2}^{T/2}\left[\frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n\cos(n\omega t) + \sum_{n=1}^{\infty} B_n\sin(n\omega t)\right]\sin(m\omega t)\mathrm{d}t$$

The three terms on the right are

$$\begin{aligned}
\frac{1}{T}\int_{-T/2}^{T/2}\frac{1}{2}A_0\sin(m\omega t)\mathrm{d}t &= 0 \\
\frac{1}{T}\int_{-T/2}^{T/2}\sum_{n=1}^{\infty} A_n\cos(n\omega t)\sin(m\omega t)\mathrm{d}t &= \sum_{n=1}^{\infty} A_n\frac{1}{T}\int_{-T/2}^{T/2}\cos(n\omega t)\sin(m\omega t)\mathrm{d}t \;=\; 0 \\
\frac{1}{T}\int_{-T/2}^{T/2}\sum_{n=1}^{\infty} B_n\sin(n\omega t)\sin(m\omega t)\mathrm{d}t &= \sum_{n=1}^{\infty} B_n\frac{1}{T}\int_{-T/2}^{T/2}\sin(n\omega t)\sin(m\omega t)\mathrm{d}t \;=\; B_m\frac{1}{2}
\end{aligned}$$

Hence, in agreement with the earlier statement,

$$\frac{1}{T}\int_{-T/2}^{T/2} f(t)\sin(m\omega t)\mathrm{d}t = B_m\frac{1}{2}$$

Similarly, to obtain A_m we would multiply by $\cos(m\omega t)$ and average over one period.

1.8 ♣ Examples

♣ **Eg 1: Square Wave** The period is T, so $\omega = 2\pi/T$.

$$A_m = \frac{2}{T}\int_{-T/2}^{+T/2} f(t)\cos(m\omega t)\mathrm{d}t = \frac{2}{T}\left[\int_{-T/2}^{0} -1\ \cos(m\omega t)\mathrm{d}t + \int_{0}^{T/2} 1\ \cos(m\omega t)\mathrm{d}t\right] = 0$$

$$\begin{aligned} B_m &= \frac{2}{T}\int_{-T/2}^{+T/2} f(t)\sin(m\omega t)\mathrm{d}t = \frac{2}{T}\left[\int_{-T/2}^{0} -1\ \sin(m\omega t)\mathrm{d}t + \int_{0}^{T/2} 1\ \sin(m\omega t)\mathrm{d}t\right] \\ &= \frac{4}{T}\int_{0}^{T/2} sin(m\omega t)\mathrm{d}t = \frac{4}{T}\frac{1}{m\omega}\left[-\cos(-m\omega t)\right|_0^{T/2} \\ &= \frac{2}{m\pi}\left[-\cos(-m\pi)+1\right] = \frac{2}{m\pi}\left[(-1)^{m+1}+1\right] \end{aligned}$$

This gives the series stated earlier:

$$f(t) = \frac{4}{\pi}\left[\sin(\omega t) + \frac{1}{3}\sin(3\omega t) + \frac{1}{5}\sin(5\omega t) + \ldots\right]$$

♣ **Eg 2: Triangular waveform** Period is 2π hence $\omega = 1$.

$$A_m = \frac{2}{2\pi}\int_{-\pi}^{+\pi} f(t)\cos(m\omega t)\mathrm{d}t = \frac{1}{\pi}\left[\int_{-\pi}^{0} -t\cos(mt)\mathrm{d}t + \int_{0}^{\pi} t\cos(mt)\mathrm{d}t\right]$$

Use integration by parts with $u = t$, $dv/dt = \cos mt$:

$$A_m = \frac{1}{\pi m^2}\left[-2 + 2\cos m\pi\right] = \frac{2}{\pi m^2}\left[-1 + (-1)^m\right] = \begin{cases} \pi & \text{for } m = 0 \\ 0 & \text{for } m \text{ EVEN} \\ -4/(\pi m^2) & \text{for } m \text{ ODD} \end{cases}$$

$$B_m = \frac{2}{2\pi}\left[\int_{-\pi}^{0} -t\sin(mt)\mathrm{d}t + \int_{0}^{\pi} t\sin(mt)\mathrm{d}t\right] = \text{grind} = 0\ .$$

$$f(t) = \frac{\pi}{2} + \frac{-4}{\pi}\left[\cos(t) + \frac{1}{9}\cos(3t) + \frac{1}{25}\cos(5t) + \ldots\right]$$

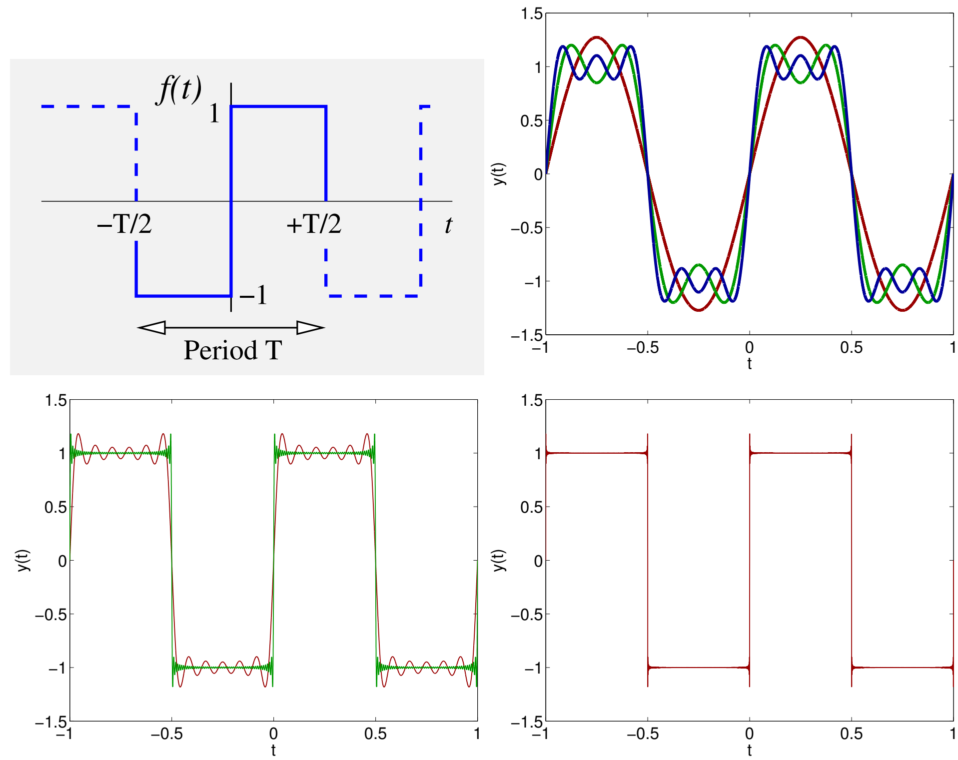


Figure 1.6: The FS of a square wave built up over 1,3,5 terms; then 11 and 101 terms; then 1001 terms. The slight "chip" at the discontinuity is a result of the Gibbs' phenomenon, discussed later.

♣ **Eg 3: Train of tophats** The period is T, so $\omega = 2\pi/T$. The function is even, so only A_n coefficients exist.

$$A_0 = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\, dt = \frac{2a}{T} \qquad\qquad A_n = \frac{4}{T}\int_0^{a/2} \cos n\omega t dt \;=\; \frac{4}{T}\frac{1}{n\omega}\sin n\omega t|_0^{a/2}$$

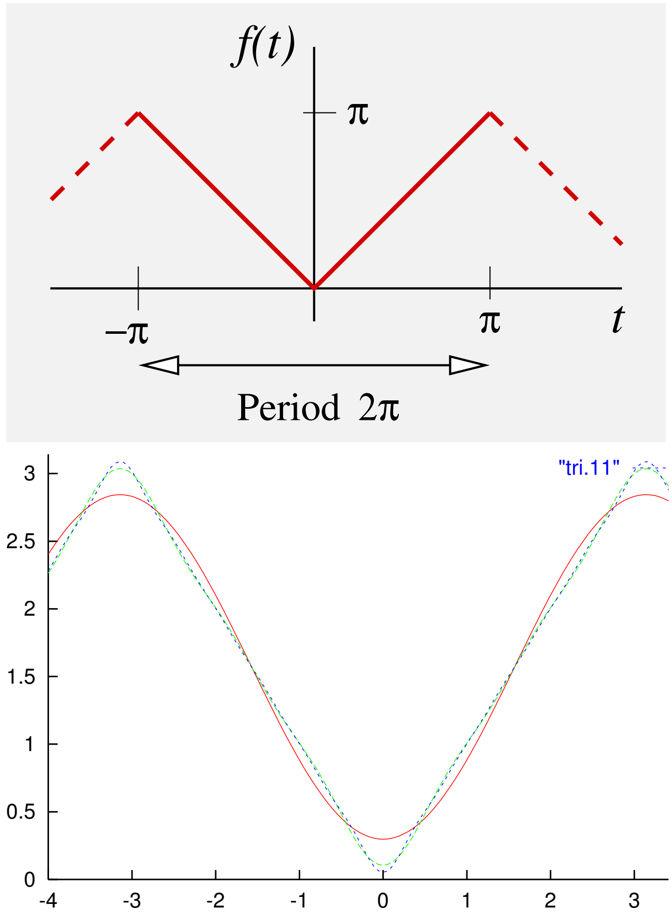
$$= \frac{2}{n\pi}\sin(n\omega a/2)$$

So $f(t) = \dfrac{a}{T} + \displaystyle\sum_{n=1}^{\infty} \frac{2}{n\pi}\sin(n\omega a/2)\cos n\omega t$

Of interest later are the values of the A_n. Suppose we set the (on/off) ratio to be $\alpha = a/T$,

$$A_n = 2\alpha\frac{\sin(n\pi\alpha)}{(n\pi\alpha)}$$

If the on/off ratio $\alpha = 1/\pi$ then the A_n are taken from the $\sin(x)/(x)$ curve as shown in Figure 1.8(b). If we reduce α, here by 1/2, then the A_n values are sampled from the curve more closely, as in Figure 1.8(c).



Using 1 then 5 then 11 terms

Figure 1.7:

1.9 Not any periodic function ...

There is a set of conditions, known as the *Dirichlet conditions*, that determine whether or not a function can be expressed as a Fourier Series. A function must

1. be periodic, or be of finite extent so that it can be made periodic by extension;
2. have only a finite number of discontinuities within a period;
3. the discontinuities must be of finite size.
4. finite number of maxima and minima

So, we can find the FS of the function $f(t) = \mathrm{e}^{-t}$, $-1 \leq t < -1$, in Figure 1.9(a), but not of $f(t) = 1/t$ for $-1 \leq t < -1$ in Figure 1.9(b).

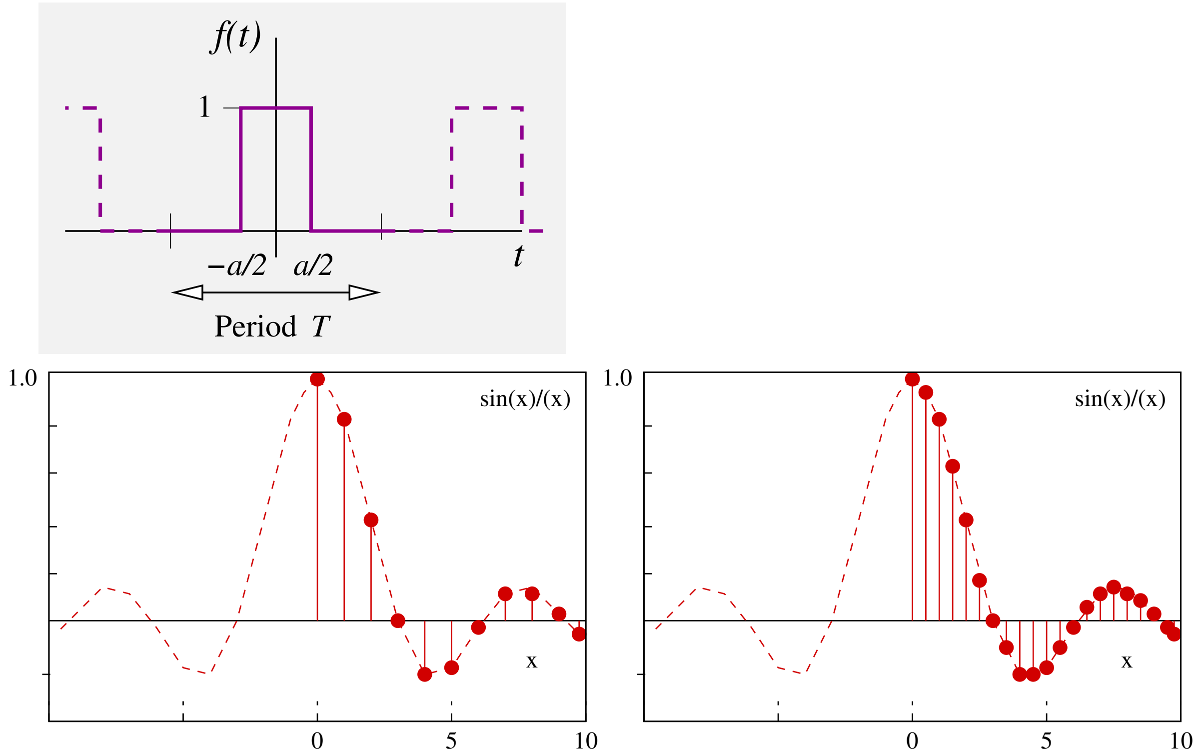


Figure 1.8:

1.10 Fourier Series at Discontinuities

Provided the Dirichlet conditions are satisfied, at a point of discontinuity in the orginal function, a Fourier series converges to

$$\mathrm{FS}(t) \rightarrow \frac{1}{2}\left(f(t_-) + f(t_+)\right)$$

where $f(t_-)$ is the value of the signal $f(t)$ just below the discontinuity, and $f(t_+)$ that just above.

We have already seen this in the square wave example. We have also seen is that we require a large number of terms in the series to faithfully reproduce the function at a discontinuity.

1.11 Symmetry properties

The task of deriving series coefficients is made a little easier by exploiting symmetries in the signal $f(t)$ and the basis functions $\sin(n\omega t)$ and $\cos(n\omega t)$.

The sine basis functions all have have odd 1/2-wave symmetry; ie, $\sin(n\omega t) = -\sin(-n\omega t)$ (Figure 1.10). If the signal $f(t)$ is even, then all integrals

$$\int_{-T/2}^{T/2} f(t)\sin(n\omega t)\mathrm{d}t$$

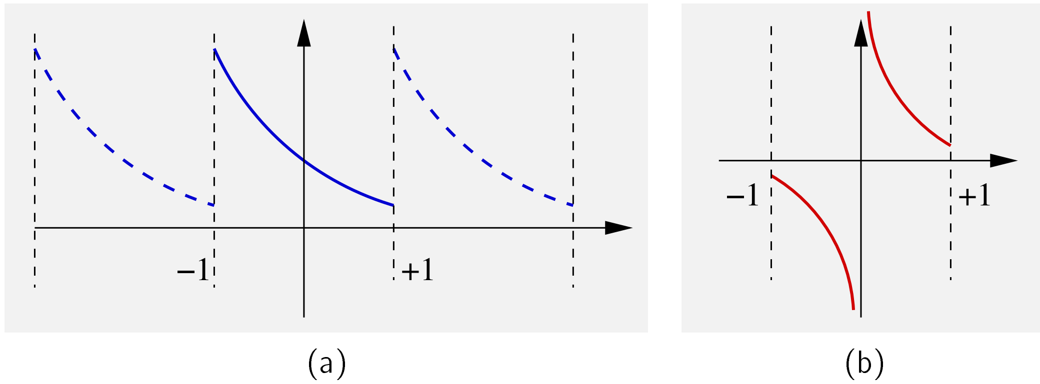


Figure 1.9: The function $1/t$ in (b) has an infinite discontinuity at $t = 0$.

must be zero. Therefore

- an even signal is contains only cosine terms; and, similarly,
- an odd signal contains only sine terms.

One also notes that for an even, then odd, signals $f(t)$

$$\int_{-T/2}^{T/2} f(t)\cos(n\omega t)\mathrm{d}t = 2\int_{0}^{T/2} f(t)\cos(n\omega t)\mathrm{d}t$$

$$\int_{-T/2}^{T/2} f(t)\sin(n\omega t)\mathrm{d}t = 2\int_{0}^{T/2} f(t)\sin(n\omega t)\mathrm{d}t$$

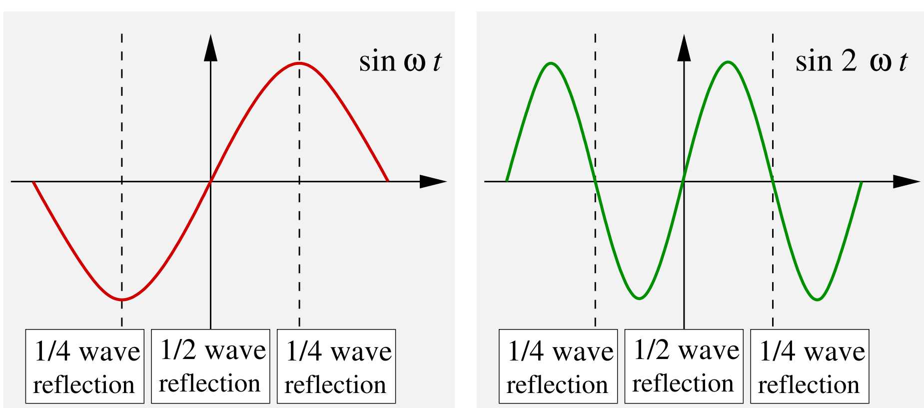
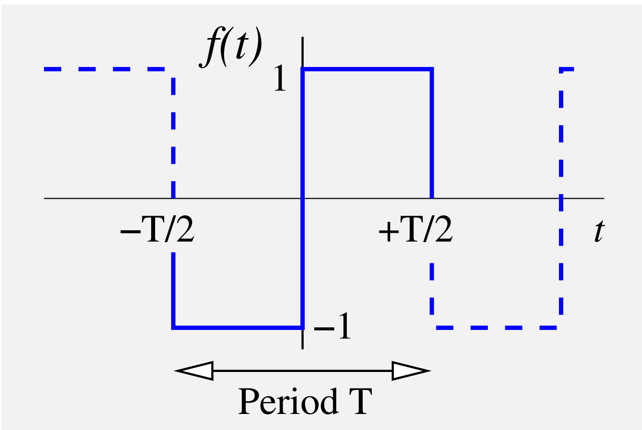
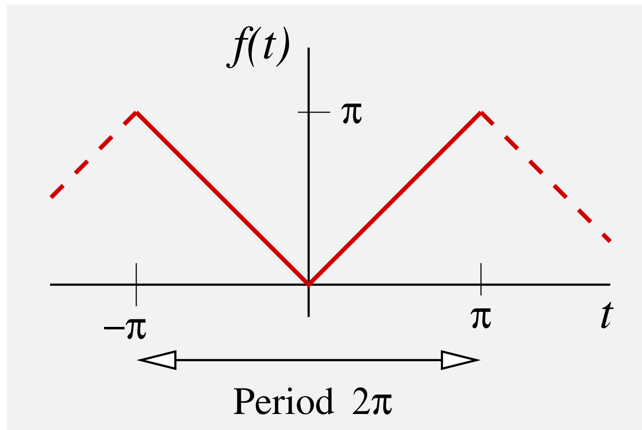


Figure 1.10: Basis function $\sin(\omega t)$ has odd 1/2-wave, even 1/4-wave symmetry. $\sin(2\omega t)$ has odd 1/2-wave, odd 1/4-wave symmetry.

Further use can be made of symmetries about the 1/4-wave points. Any $\sin(n\omega t)$ with n-even has odd symmetry about these points. Thus if a signal $f(t)$ has even symmetry about the 1/4-wave points, any n-even sine terms will vanish. However if the signal's symmetry is odd about the 1/4-wave points, the n-odd sine terms vanish. Similar arguments can be made for the cosine series. The square wave and triangular waves (Figure 1.11) are examples, but do note that the signal $f(t)$ does *not* have to have 1/2-wave symmetry to exploit 1/4-wave symmetry. One could consider higher symmetries, but they get increasingly difficult to recognize.



$\frac{4}{\pi}\left[\sin(\omega t)+\frac{1}{3}\sin(3\omega t)+\frac{1}{5}\sin(5\omega t)+\cdots\right]$



$\frac{\pi}{2}+\frac{-4}{\pi}\left[\cos(t)+\frac{1}{9}\cos(3t)+\frac{1}{25}\cos(5t)+\ldots\right]$

Figure 1.11:

1.12 Completing Functions

Suppose you need derive the Fourier series of a non- time-continuous, non-period function. For example, $f(t) = t$ for $0 \le t \le 1$. First, you MUST make the function periodic — but exactly how is a matter of choice. Figure 1.12 shows three possible ways of completing the example function.

Which is best? Because the cosine series has no discontinuities it will require fewer terms to make an overall decent approximation. However, the kink at $t = 0$ will not be accurate. If a good approximation with few terms is required close to $t = 0$, it is probably best to use the sine series completion.

1.13 The Gibbs phenomenon

Something rather curious occurs in the FS at a discontinuity. Residual oscillations lead to an overshoot, whose size is a characterstic of the underlying function. (It can be large! For a square wave of amplitude A, the overshoot is around $\delta = 0.18A$.) As more and more terms are added to the series, the oscillations get squeezed into a shorter and shorter region around the discontinuity, *but their characteristic amplitude remains constant*! This effect is known as the Gibbs phenomenon.

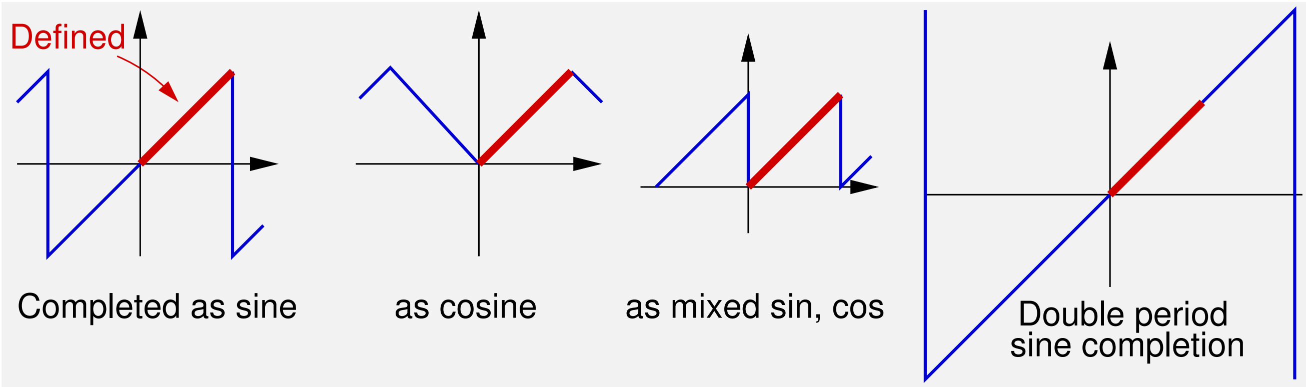


Figure 1.12:

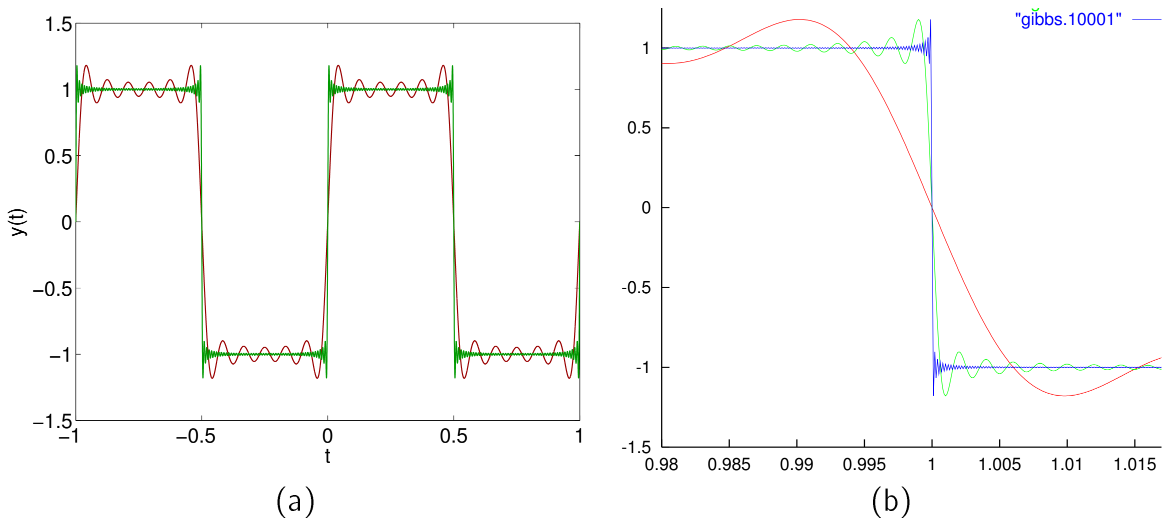


Figure 1.13: The Gibbs phenomenon. (b) shows the discontinuity of (a) at a finer time scale, but with more terms added.

1.14 Mean square values of Fourier Series Parseval's Theorem

The instantaneous power in a signal is proportional to the its modulus squared. For a periodic signal we can derive the **average signal power** by integrating over a period.

$$\begin{aligned}
\textbf{Ave signal pwr} &= \frac{1}{T}\int_{-T/2}^{+T/2} |f(t)|^2 \mathrm{d}t \\
&= \frac{1}{T}\int_{-T/2}^{+T/2} \left\{ \left| \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega t) + \sum_{n=1}^{\infty} B_n \sin(n\omega t) \right|^2 \right\} dt
\end{aligned}$$

At first this looks nightmarish, because the squaring introduces an infinite number of cross terms, on the bottom line of the next expression.

$$\frac{1}{T}\int_{-T/2}^{+T/2}\left\{\left(\frac{1}{2}A_0\right)^2 + A_1^2\cos^2\omega t + A_2^2\cos^2 2\omega t + \ldots + B_1^2\sin^2\omega t + B_2^2\sin^2 2\omega t + \ldots + \right.$$
$$\left. A_0A_1\cos\omega t + \ldots + A_0B_1\sin\omega t + \ldots + 2A_1B_1\cos\omega t\sin\omega t + 2A_1B_2\cos\omega t\sin 2\omega t + \ldots\right\}\mathrm{d}t$$

However, when you integrate over cross terms, orthogonality will get rid of all the terms on the bottom line! So, we are left with the simple result that the mean square is

$$\boxed{\textbf{Ave pwr} = \frac{1}{T}\int_{-T/2}^{+T/2}|f(t)|^2\mathrm{d}t = \left(\frac{1}{2}A_0\right)^2 + \frac{1}{2}\sum_{n=1}^{\infty}(A_n)^2 + \frac{1}{2}\sum_{n=1}^{\infty}(B_n)^2}$$

A good way to remember this is as

$$\textbf{Mean Square} = (\text{d.c. amplitude})^2 + \frac{1}{2}\sum(\text{a.c. amplitude})^2$$

which is true whether the signal is pure d.c. or single frequency a.c.

The Root mean square value of a periodic signal is therefore

$$\textbf{Root Mean Square} = \sqrt{\left(\frac{1}{2}A_0\right)^2 + \frac{1}{2}\sum_{n=1}^{\infty}(A_n)^2 + \frac{1}{2}\sum_{n=1}^{\infty}(B_n)^2}$$

1.15 ♣ Fourier and Parseval in an electrical circuit

[Q] Suppose the full wave rectified voltage in Figure 1.14(a) is applied to the circuit in (b). What is the average power dissipated?

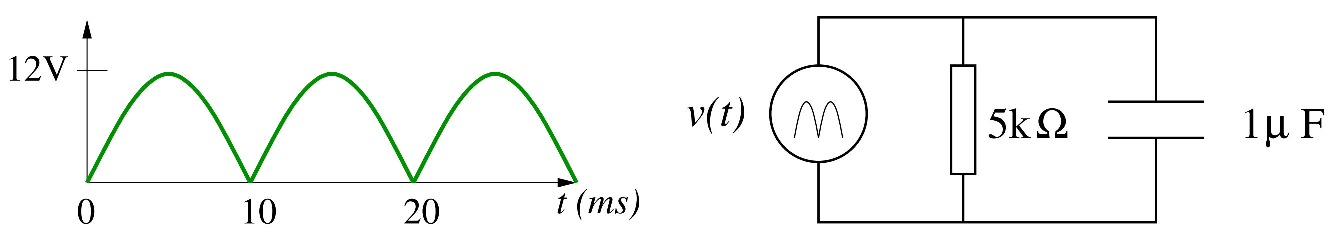


Figure 1.14:

[A] From the diagram $T = 10^{-2}\,\mathrm{s}$ and $\omega_0 = 2\pi/T = 200\pi$. By grinding, or from HLT, we find the signal $v(t)$, with t in seconds, is

$$v(t) = \frac{24}{\pi}\left\{1 + \frac{2}{3}\cos\omega_0 t - \frac{2}{15}\cos 2\omega_0 t + \ldots\right\}$$

We know that power is dissipated in the resistor alone, so

$$P_{\text{ave}} = \frac{1}{R}\overline{v^2} = \frac{1}{5\times 10^3}\left(\frac{24}{\pi}\right)^2\left\{1^2 + \frac{1}{2}\left(\left(\frac{2}{3}\right)^2 + \left(\frac{2}{15}\right)^2 + \ldots\right)\right\}$$

[Q] What is current drawn from the source?

The current drawn from the source at a single frequency ω is $I(\omega) = Y(\omega)V(\omega)$, where Y is the admittance. But we have voltage components at $\omega = 0$, $\omega = \omega_0$, $\omega = 2\omega_0$ and so on. We must work out the admittances at all these frequencies.

$$\begin{aligned} Y(\omega) &= \frac{1}{R} + j\omega C = (2\times 10^{-4}) + j\omega(1\times 10^{-6}) \\ &= 10^{-4}\left(2 + j\frac{\omega}{100}\right) = \begin{cases} 10^{-4}(2+j\,0), & \text{dc} \\ 10^{-4}(2+j\,2\pi), & \omega = \omega_0 = 200\pi \\ 10^{-4}(2+j\,4\pi), & \omega = 2\omega_0 = 400\pi \\ 10^{-4}(2+j\,2n\pi), & \omega = n\omega_0 \end{cases} \end{aligned}$$

To avoid mixing functions of time with phasors, it makes sense to rewrite $v(t)$ as the real part of

$$v(t) = \mathrm{Re}\left(\frac{24}{\pi}\left\{1 + \frac{2}{3}e^{j\omega_0 t} - \frac{2}{15}e^{j2\omega_0 t} + \ldots\right\}\right)$$

Hence

$$\begin{aligned} i(t) &= \mathrm{Re}\left(\frac{24\times 10^{-4}}{\pi}\left\{2 + (2+j2\pi)\frac{2}{3}e^{j200\pi t} - (2+j4\pi)\frac{2}{15}e^{j400\pi t} + \ldots\right\}\right) \\ &= \frac{48\times 10^{-4}}{\pi}\left\{1 + (1+\pi^2)^{1/2}\frac{2}{3}\cos(200\pi t + \phi_1)\right. \\ &\qquad\qquad \left. -(1+4\pi^2)^{1/2}\frac{2}{15}\cos(400\pi t + \phi_2) + \ldots\right\} \end{aligned}$$

with $\phi_1 = \tan^{-1}\pi$ and $\phi_2 = \tan^{-1}2\pi$.

1.16 The Complex Fourier Series

You will have noticed while working out $i(t)$ that there was a rather unsatisfactory moment when we had to rewrite the Fourier series using an exponential representation. It raises the following question.

> **Would it be possible use an exponential, or complex, form of the Fourier Series from the outset?**

It turns out — perhaps not surprisingly — that the set $e^{in\omega t}$ provides a set of orthogonal basis functions, and that a periodic function that satisfies the Dirichlet conditions and has period $T = 2\pi/\omega$ can be written as

The complex Fourier series (NB: n ranges from $-\infty$ to $+\infty$.)

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega t}$$

The inner product is defined as integration over a period with the complex conjugate, divided by the period, and the orthogonality conditions are simpler than before,

$$\frac{1}{T}\int_{-T/2}^{T/2} e^{in\omega t} e^{-im\omega t} dt = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

We determine the form of C_n using the orthogonality relationship in either the short hand or long hand ways.

The short hand way says

$$C_m = \frac{\langle f(t) \cdot e^{im\omega t} \rangle}{\langle e^{im\omega t} \cdot e^{im\omega t} \rangle} = \frac{\frac{1}{T}\int_{-T/2}^{T/2} f(t) e^{-im\omega t} dt}{\frac{1}{T}\int_{-T/2}^{T/2} e^{im\omega t} e^{-im\omega t} dt} = \frac{1}{T}\int_{-T/2}^{T/2} f(t) e^{-im\omega t} dt$$

The long hand version is not much longer.

$$\begin{aligned}
\text{If} \quad f(t) &= \sum_{n=-\infty}^{\infty} C_n \mathrm{e}^{\mathrm{i}n\omega t} \\
\frac{1}{T}\int_{-T/2}^{T/2} f(t)\mathrm{e}^{-\mathrm{i}m\omega t}\mathrm{d}t &= \frac{1}{T}\int_{-T/2}^{T/2} \sum_{n=-\infty}^{\infty} C_n \mathrm{e}^{\mathrm{i}n\omega t}\mathrm{e}^{-\mathrm{i}m\omega t}\mathrm{d}t \\
&= \sum_{n=-\infty}^{\infty} C_n \frac{1}{T}\int_{-T/2}^{T/2} \mathrm{e}^{\mathrm{i}n\omega t}\mathrm{e}^{-\mathrm{i}m\omega t}\mathrm{d}t \\
&= C_m
\end{aligned}$$

The coefficients in the complex Fourier series are

$$C_m = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\mathrm{e}^{-\mathrm{i}m\omega t}\mathrm{d}t$$

There is a third way of deriving the complex Fourier series. It is a bit feeble because it simply uses the relationship

$$\mathrm{e}^{-\mathrm{i}m\omega t} = \cos m\omega t - \mathrm{i}\sin m\omega t \ .$$

The relationship between the C coefficients and those for the non-complex Fourier series is

$$C_m = \begin{cases} (A_m - \mathrm{i}B_m)/2 & \text{for} \quad m > 0 \\ A_0/2 & \text{for} \quad m = 0 \\ (A_{|m|} + \mathrm{i}B_{|m|})/2 & \text{for} \quad m < 0 \end{cases}$$

Also note that $C_m = C^*_{-m}$, where $*$ denotes complex conjugate.

1.17 Parseval and the Complex Fourier Series

This is asked as a question on the first tutorial sheet, and I want you to think about it, rather than copy it down.

Notes on this will appear later in the course to check your answers!

1.18 Summary

We have defined certain terms used to describe signals.

We have reviewed how periodic signals can be represented as Fourier Series — linear sums of pure harmonic signals — and revised their properties.

We have introduced the complex Fourier Series, which is often a more convenient representation to use when having to deal with phase shifts.

Interesting and useful as the Complex Fourier Series is, there is nothing in it that addresses the gap in our knowledge, which is

> how to cope with non-periodic signals of infinite duration; that is, those which have continuous spectra in the frequency domain.

This is where we move next.